21 Jackson is trying to find the density, in g/cm^3, of a block of wood.
The block of wood is in the shape of a cuboid.

He measures
the length as 13.2 cm, correct to the nearest mm
the width as 16.0 cm, correct to the nearest mm
the height as 21.7 cm, correct to the nearest mm

He measures the mass as 1970 g, correct to the nearest 5 g.

By considering bounds, work out the density of the wood.
Give your answer to a suitable degree of accuracy.

You must show all your working and give a reason for your final answer.

Quantity	Lower bound	Upper bound
Length cm	13.15	13.25
Width cm	15.95	16.05
Height cm	21.65	21.75
Mass g	1967.5	1972.5

1 mark
Half a millimetre = 0.05 cm; half of 5 g = 2.5 g

Volume lower bound = 13.15 × 15.95 × 21.65 = 4540.925125 cm^3

Volume upper bound = 13.25 × 16.05 × 21.75 = 4625.409375 cm^3

1 mark
Multiply corresponding length bounds for a volume bound

Small mass / large volume

Lower density

Volume cm^3	Mass g
4625.409375	1967.5 1967.5
1	4625.409375

Lower ≈ 0.4253677546

1 mark
Mass per 1 cm^3 gives density

Large mass / small volume

Upper density

Volume cm^3	Mass g
4540.925125	1972.5 1972.5
1	4540.925125

Upper ≈ 0.4343828506

1 mark
Both density bounds correct

Both round to 0.43 g/cm^3 at 2 d.p. (2 s.f.).

1 mark
Choose their shared precision

(Total for Question 21 is 5 marks)

TOTAL FOR PAPER IS 80 MARKS

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA